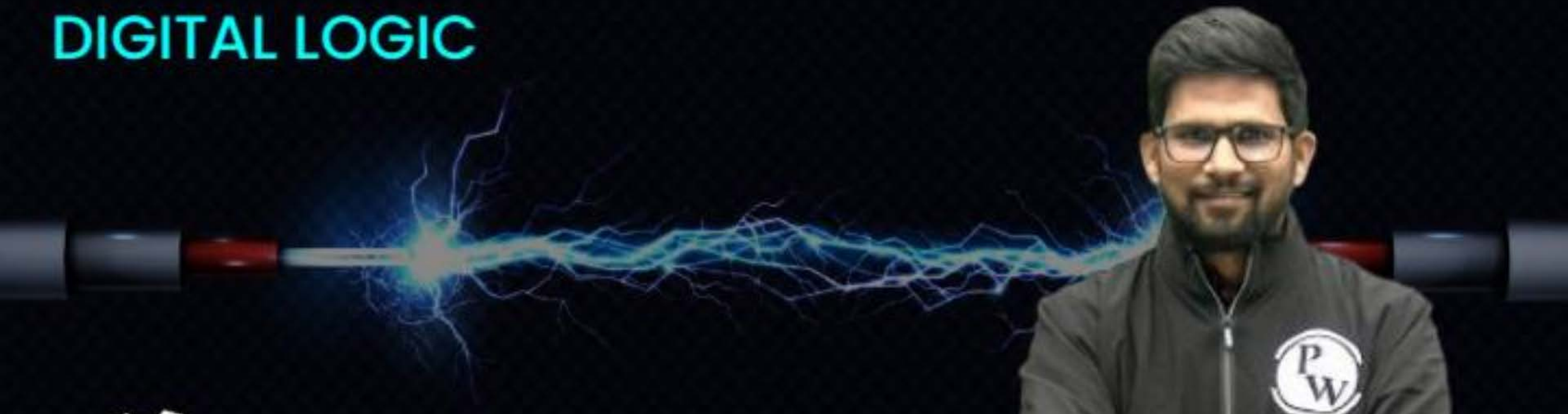
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 09

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture



- Synch Counter
-
-



Synch. Counter

- To design the counter

1 - 2 - 3 - 1 - 2 - 4 - 1 - 2 - 5 - 1 - 2 - 6 - 1 - 2 - 7 - 1 - 2 - 8

No. of ffs required is 7 .

1000 → 4ff

3-ffs → for 6 different 1's

or

6 different 2's

Total 7 ffs

Synchronous Series Carry Counter

4-bit synchronous series carry counter :

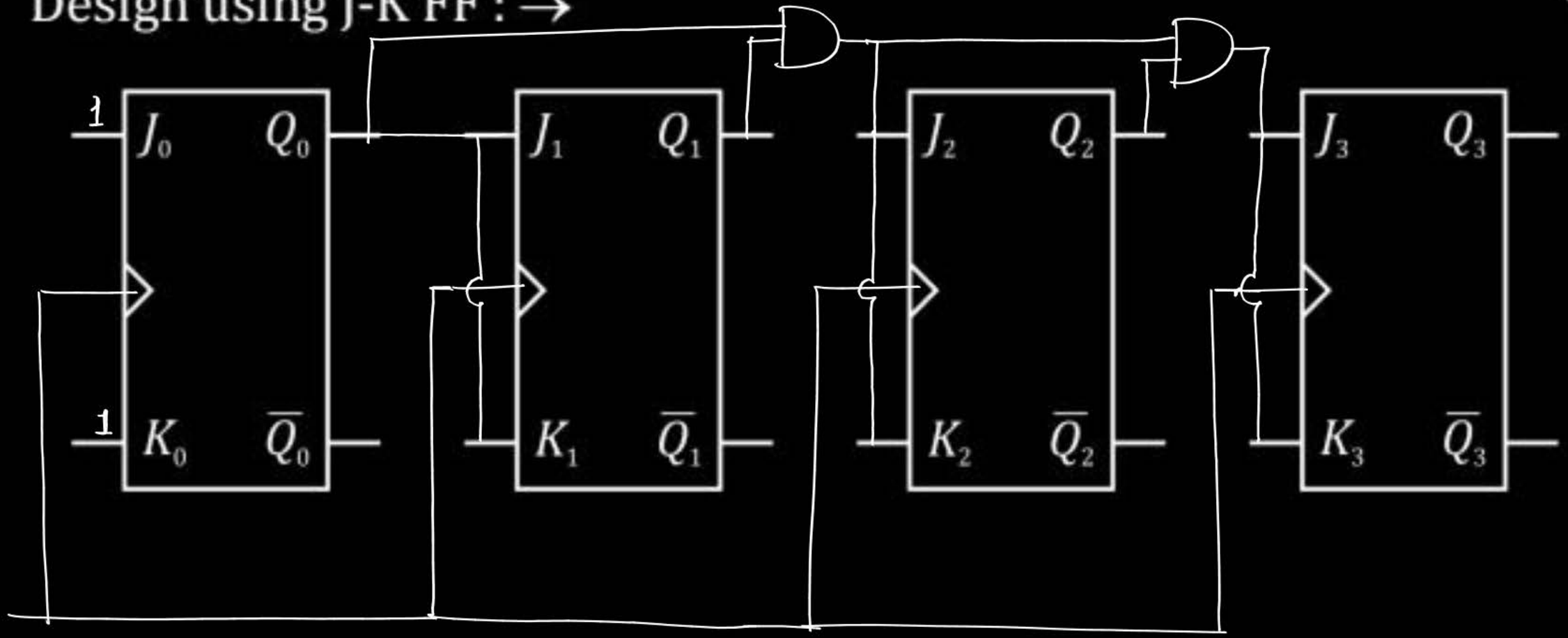
Lest analyze the sequence and design it :

Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0



Q_3	Q_2	Q_1	Q_0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1
0	0	0	0

Design using J-K FF : →

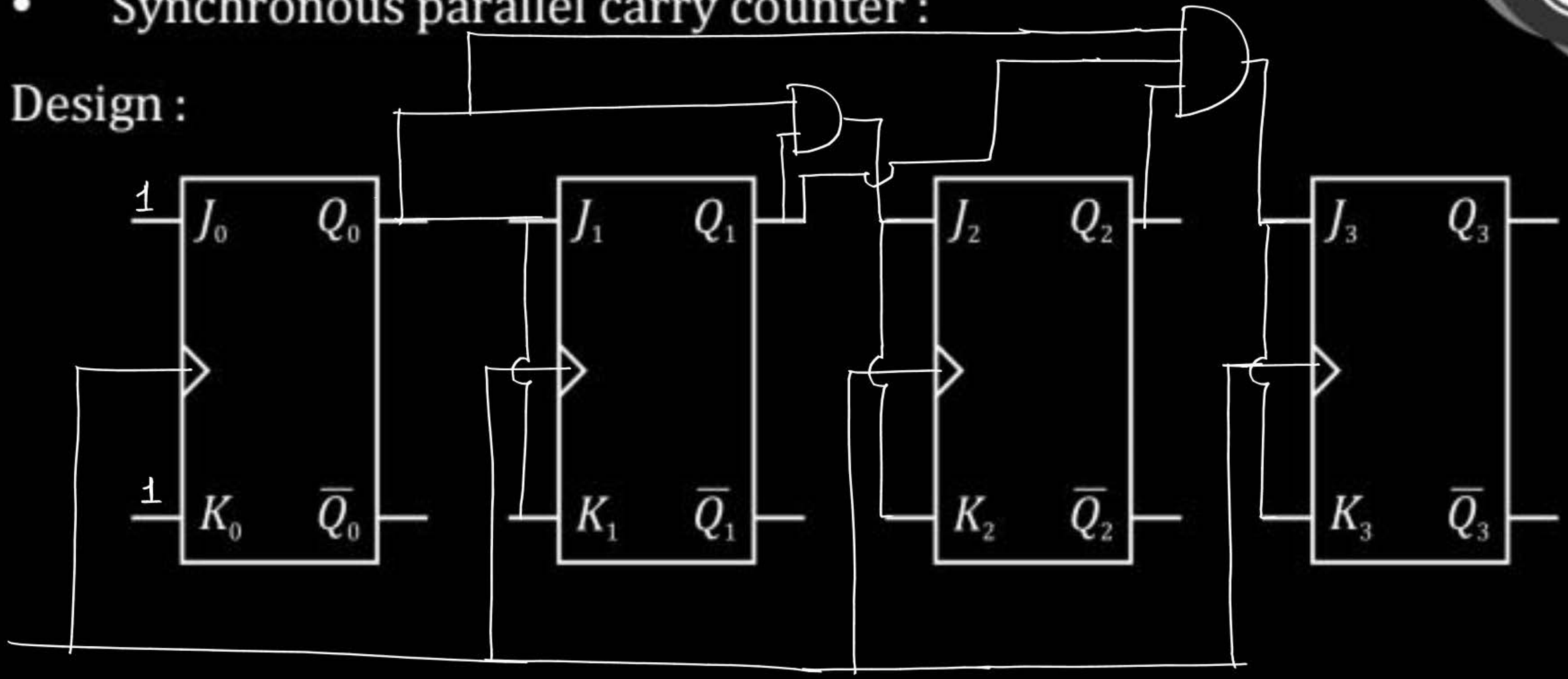


No. of AND gate require is :

for n-bit synchronous Series carry counter (n-2) [2-i/p] AND gates will be required.

- Synchronous parallel carry counter :

Design :



- AND gates here are operating in parallel with use of multi i/p – AND gates.
- No. of AND gates required is $(n-2)$ for n – bit synchronous parallel carry Counter.
  - Highest i/p AND gate → $(n-1)$ i/p

⇒ Down Counter → H.W.

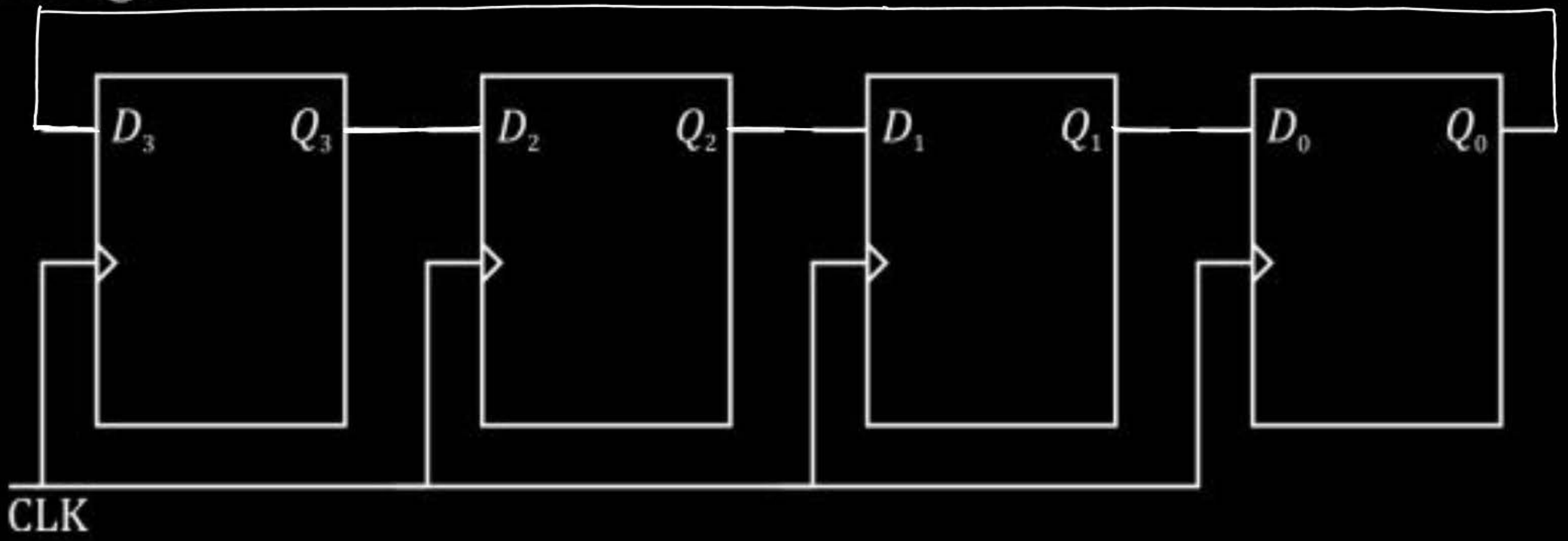
Ring Counter

$Q_3(n+1) = D_3 = Q_0(n)$, $Q_2(n+1) = D_2 = Q_3(n)$

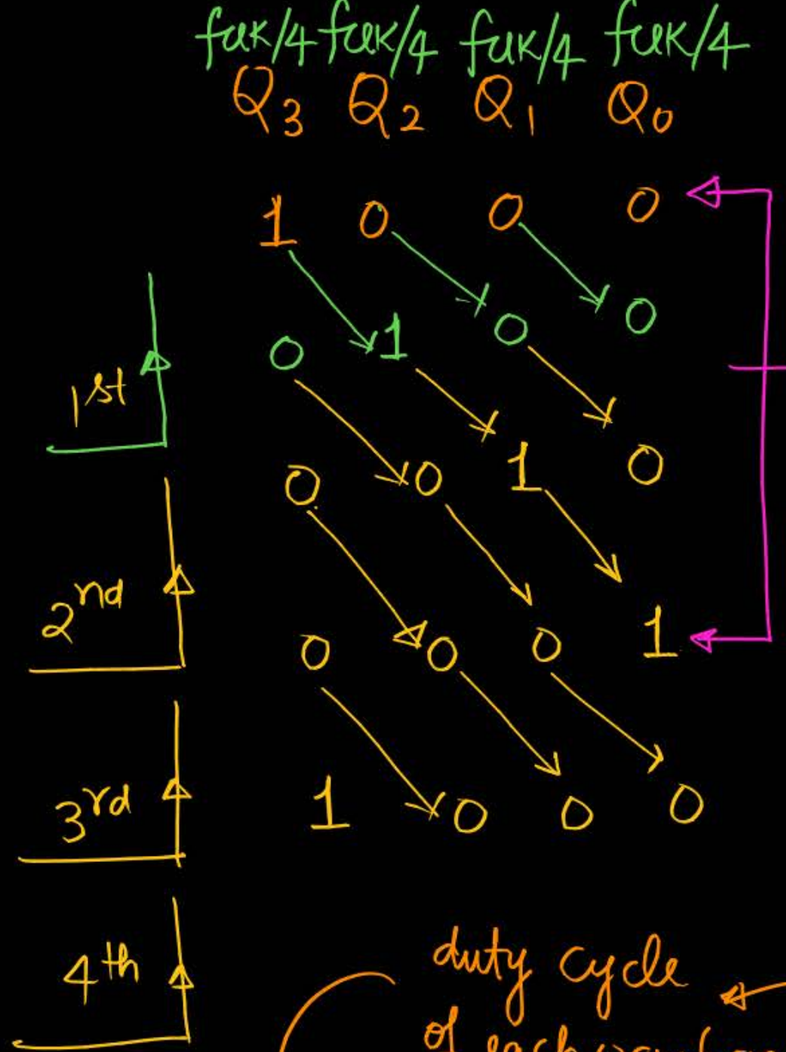
$Q_1(n+1) = D_1 = Q_2(n)$, $Q_0(n+1) = D_0 = Q_1(n)$

4-bit Ring Counter :

Circuit Design :



Working :



n-bit ring Counter then at every o/p

→ $T_{off} = (n-1)\,T_{clk}$

$T_{ON} = 1\,T_{clk}$

$T_Q = nT_{clk} = T_{off} + T_{ON}$

$f_Q = f_{clk}/n$.

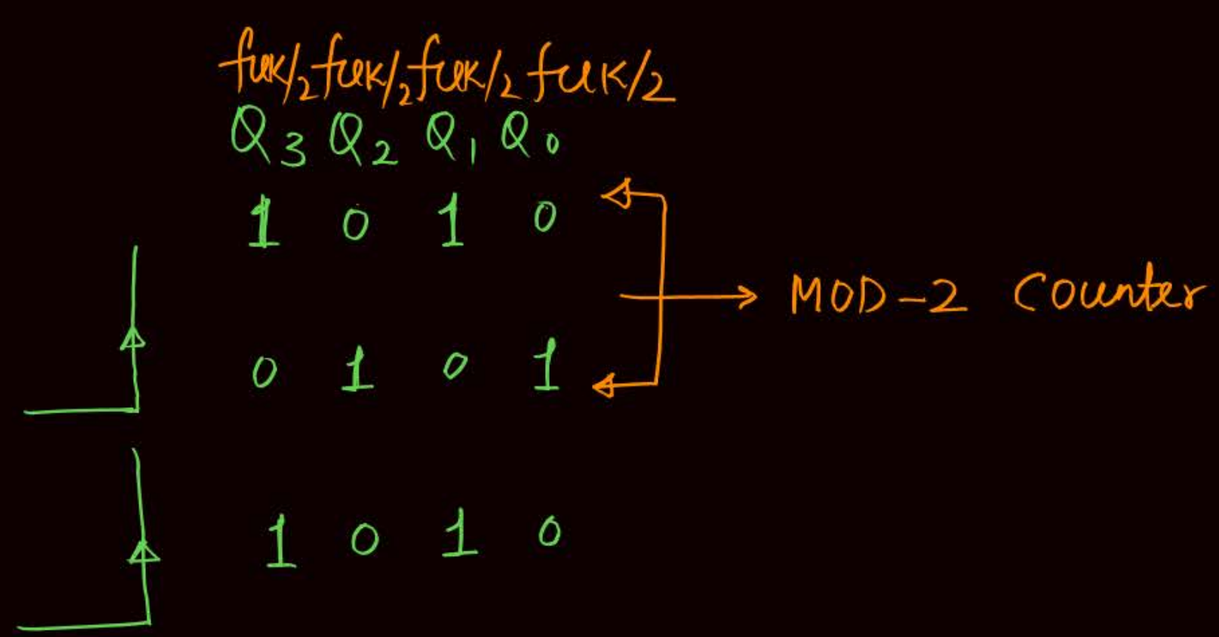
duty cycle of each waveform

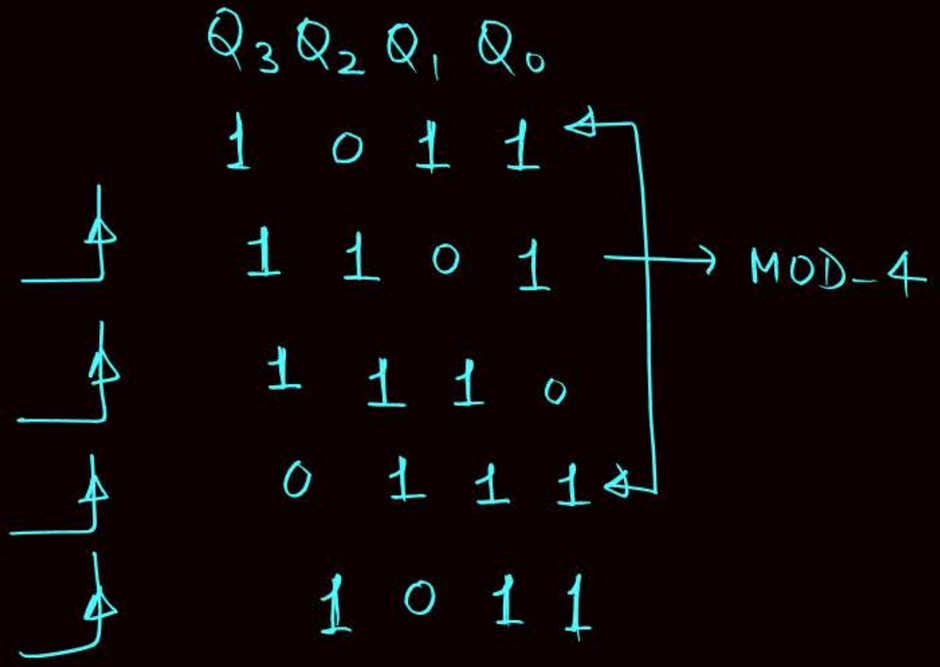
% duty cycle

$= \frac{T_{clk}}{nT_{clk}} \times 1 = 1/n \times 100\%$

Imp points of Ring Counter : →

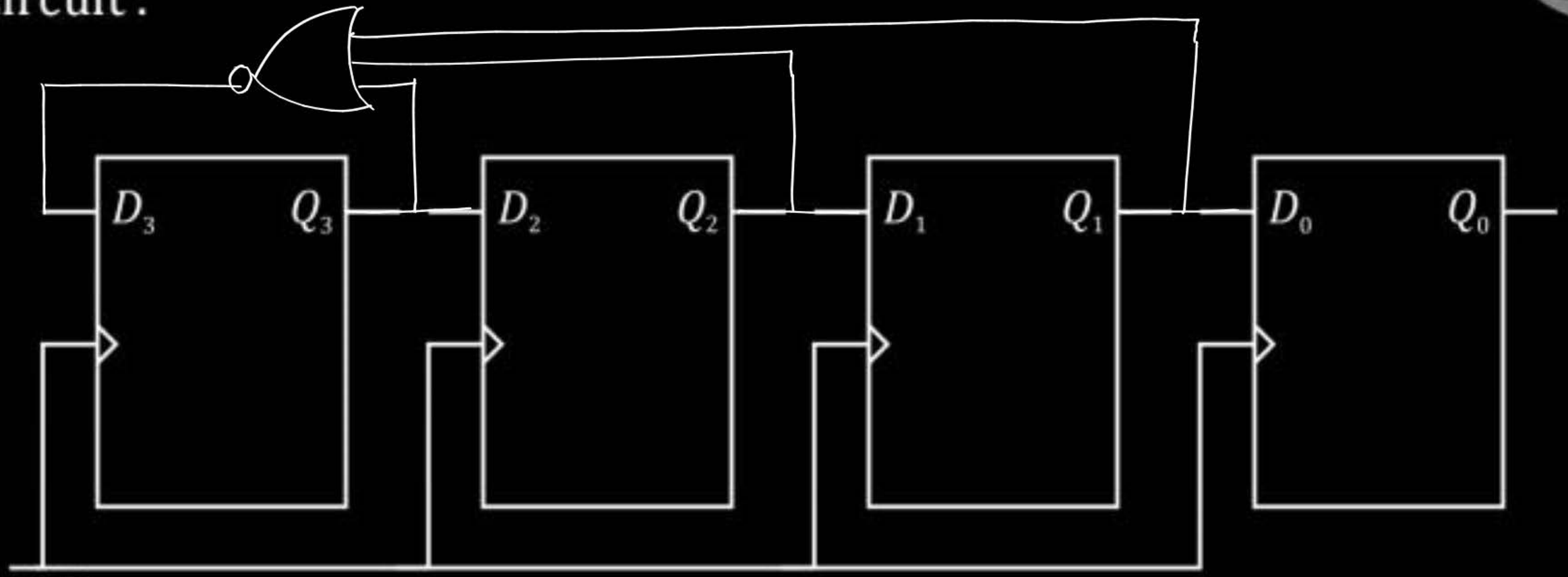
- MOD no of the n-bit ring Counter is n.
- At every o/p input clock frequency is divided by MOD no. of the counter (f_{CLK}/n).
- Unused states in Ring Counter $= (2^n - n)$
- Duty cycle of each o/p waveform will be $1/n \times 100\%$.


fclk/2 fclk/2 fclk/2 fclk/2
Q3 Q2 Q1 Q0
1 0 1 0
0 1 0 1
1 0 1 0
MOD-2 Counter


Q3 Q2 Q1 Q0
1 0 1 1
1 1 0 1
1 1 1 0
0 1 1 1
1 0 1 1
MOD-4

Self Starting Ring Counter Circuit

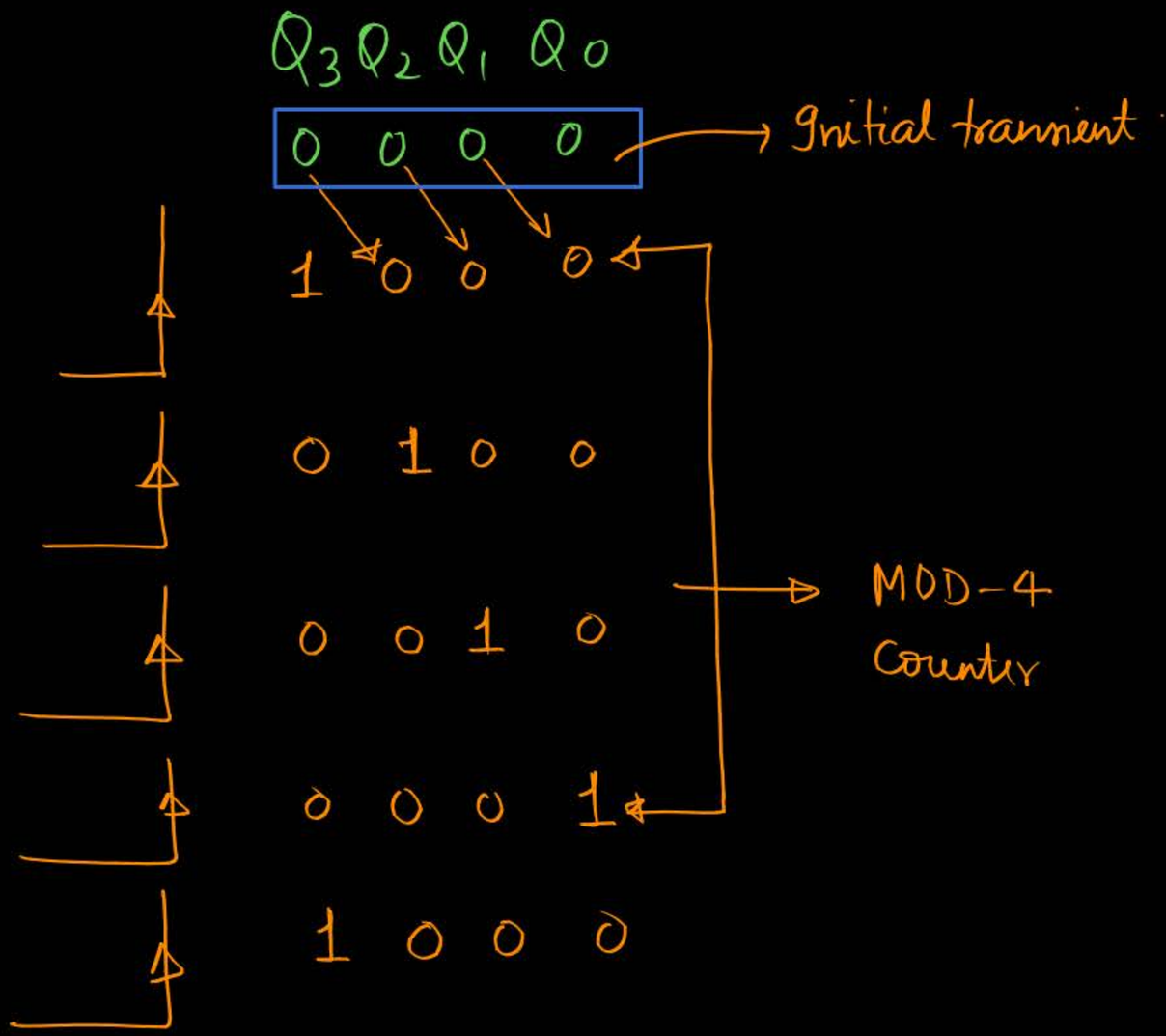
Circuit :



$Q_2(n+1) = Q_3(n)$, $Q_1(n+1) = Q_2(n)$, $Q_0(n+1) = Q_1(n)$

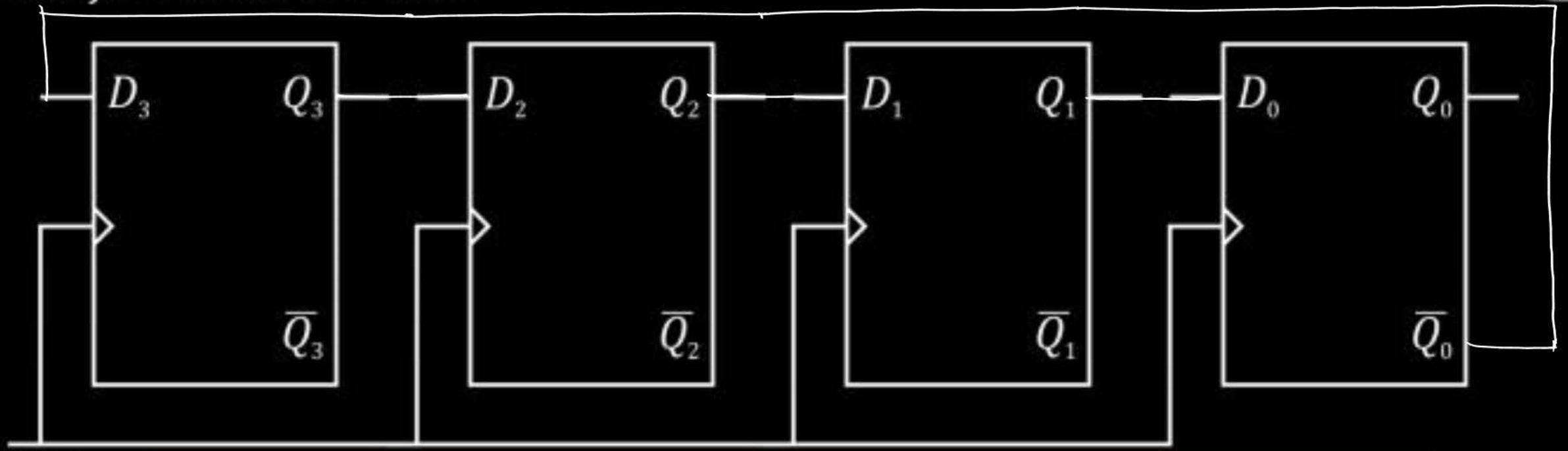
$Q_3(n+1) = D_3 = \overline{Q_3(n) + Q_2(n) + Q_1(n)}$

Working :


$Q_3 Q_2 Q_1 Q_0$
0 0 0 0
Initial transient
1 0 0 0
0 1 0 0
0 0 1 0
0 0 0 1
1 0 0 0
MOD-4
Counter

Johnson Counter or Twisted Ring Counter

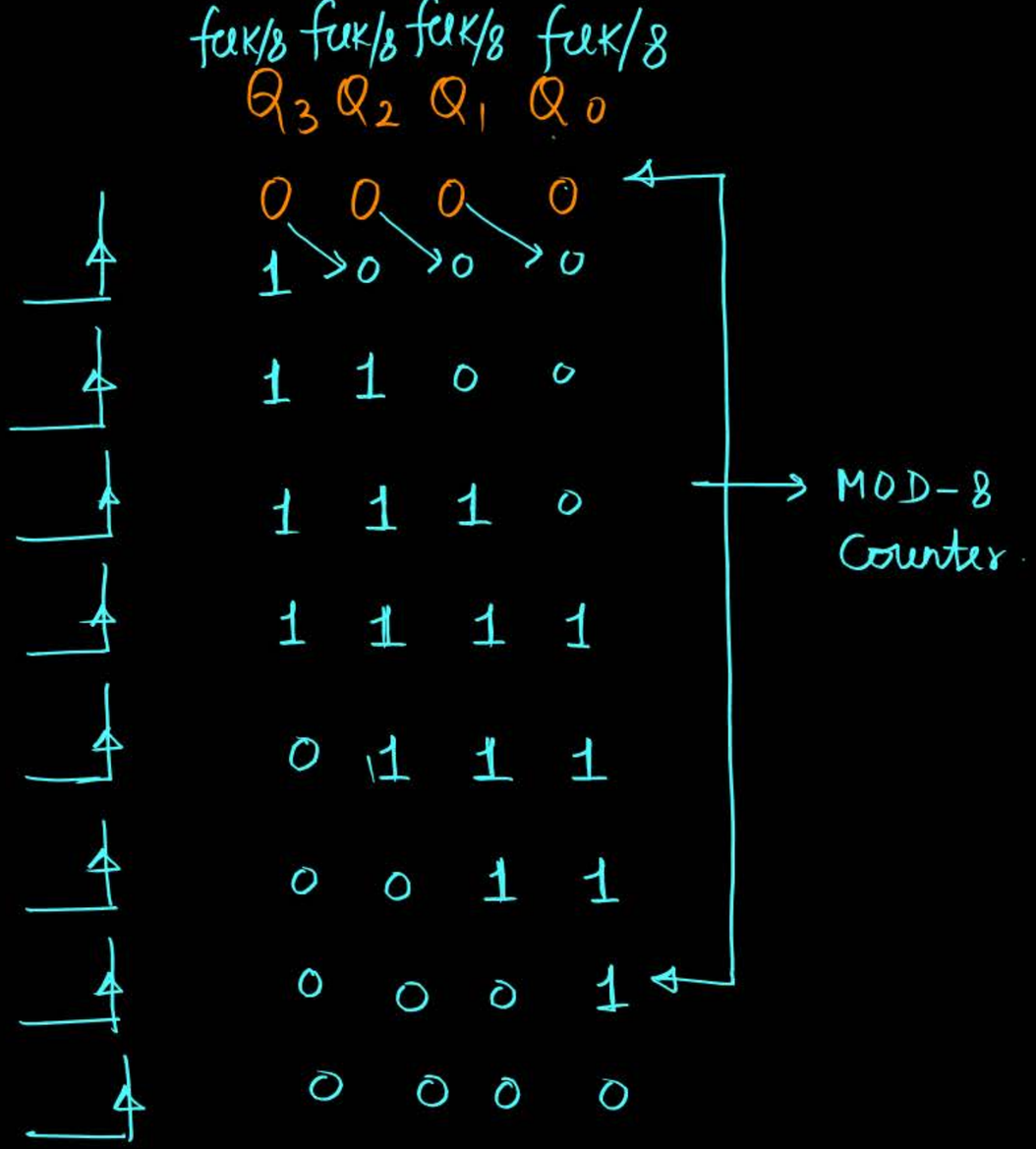
4-bit Johnson Counter :



$Q_2(n+1) = D_2 = Q_3(n)$, $Q_1(n+1) = D_1 = Q_2(n)$, $Q_0(n+1) = D_0 = Q_1(n)$

$Q_3(n+1) = D_3 = \overline{Q_0}(n)$

Working :



n-bit Johnson Counter

at each ff o/p

$T_{off} = nT_{clk}$

$T_{ON} = nT_{clk}$

$T_Q = 2nT_{clk}$

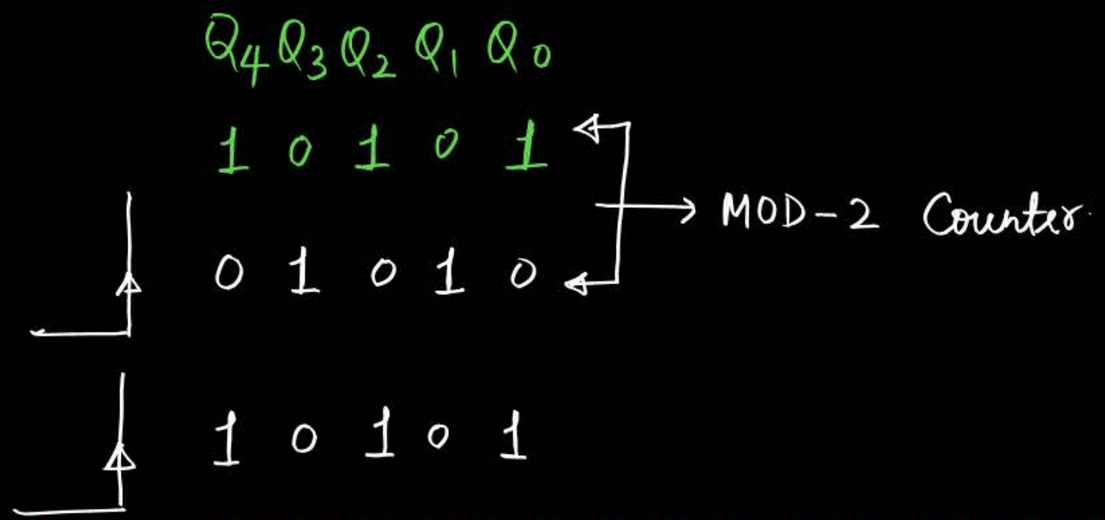
$f_Q = \frac{f_{clk}}{2n}$

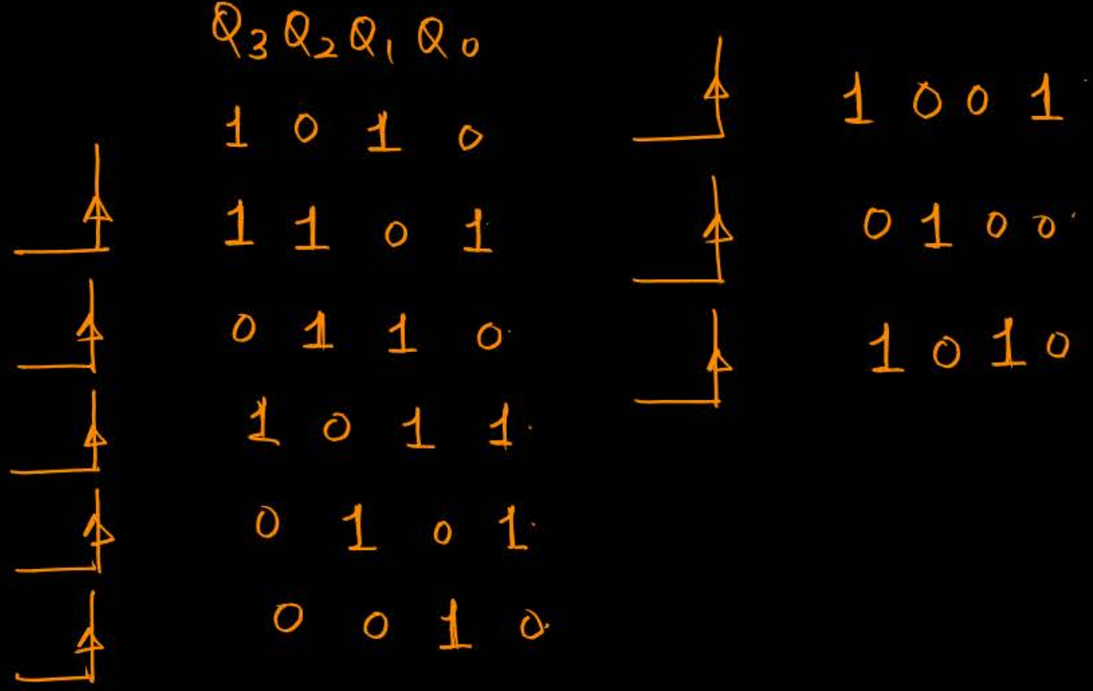
% duty cycle at each o/p $= \frac{nT_{clk}}{2nT_{clk}} \times 100$

$= 50\%$

Imp points of Johnson Counter : →

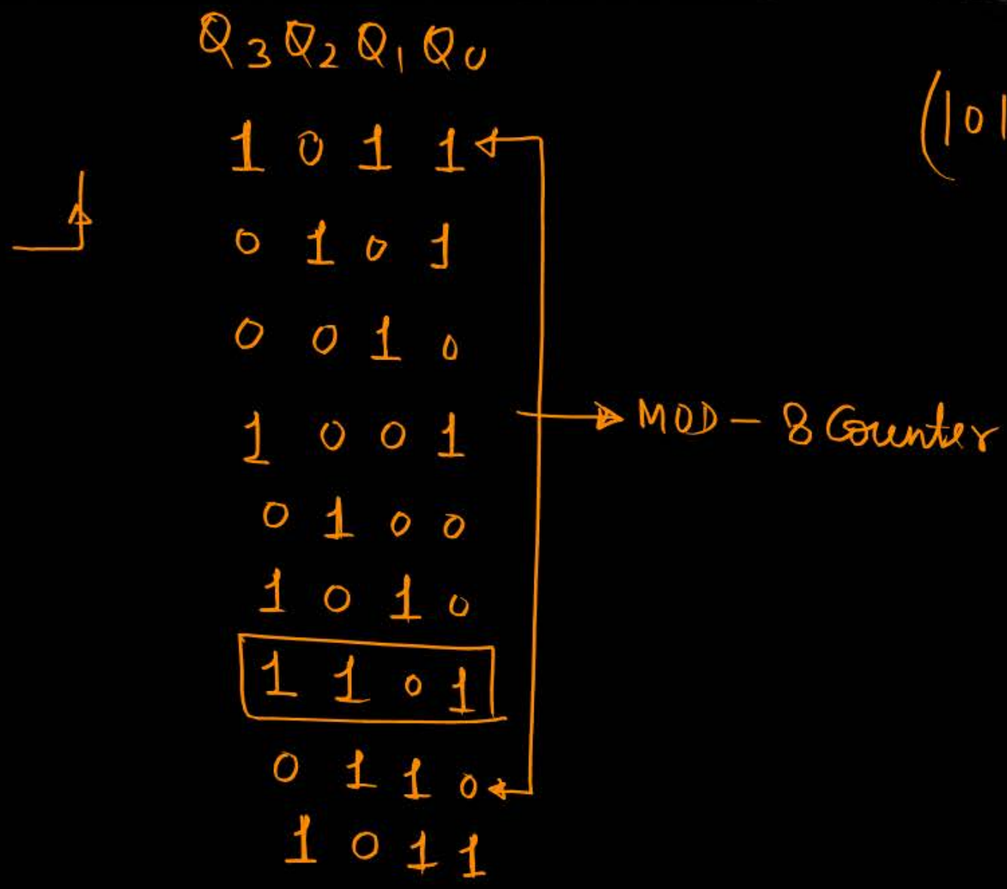
- MOD no. of Johnson Counter designed using n-FFs is 2n.
- At each ff O/P, input clock frequency is divided by the MOD no. [fclk/2n].
- No. of unused states = $(2^n - 2n)$
- Duty cycle of the waveform at each ff O/P is 50% and independent of no. of ffs used.


$Q_4 Q_3 Q_2 Q_1 Q_0$
1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
MOD-2 Counter


$Q_3 Q_2 Q_1 Q_0$
1 0 1 0
1 1 0 1
0 1 1 0
1 0 1 1
0 1 0 1
0 0 1 0
1 0 0 1
0 1 0 0
1 0 1 0
MOD-8

Question

We have 4-bit Johnson counter. Its starting state is $(1011)_2$. Then it will be in what state after 22 CLK pulses (13)$_{10}$.



$(1011)_2 \xrightarrow[16\text{ clk}]{2\times 8} (1011)_2 \xrightarrow{6\text{ clk}} (1101)_2$

$= (13)_{10}$

Question

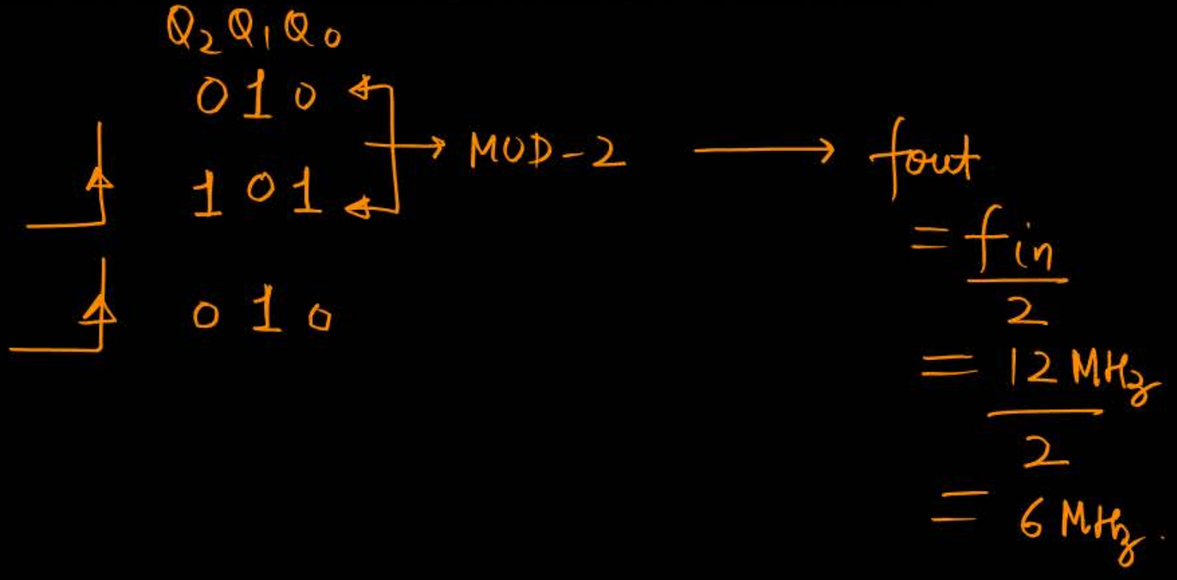
We have 3-bit Johnson counter whose starting state is $(010)_2$. If input clock frequency 12 MHz then value of output frequency will be

(a) 6 MHz

(b) 2 MHz

(c) 4 MHz

(d) 8 MHz



Question MSQ

Which of the following is/are true ? (A, C)

(a) In ring counter duty cycle of the output waveform decreases with increase in the no. of FFs. % duty cycle = 1/n × 100%

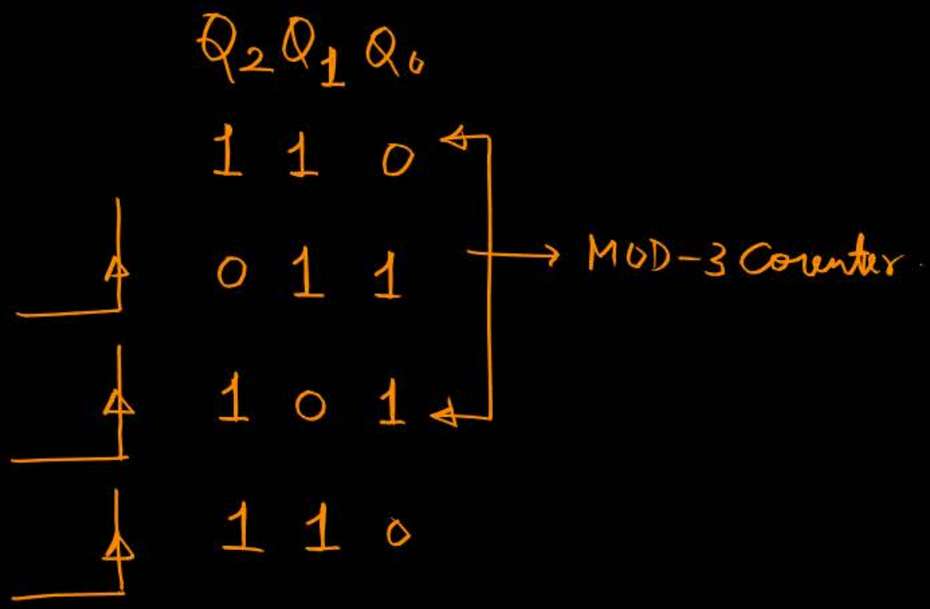
(b) Johnson counter has duty cycle of 1/2n, where n is no. of FFs. ×

(c) No. of unused states in ring counter is $(2^n - n)$.

(d) No. of unused states in Johnson counter is $2^{(n-1)}$. × $(2^n - 2n)$

Question

In a 3-bit ring counter with starting state $Q_2Q_1Q_0 = (110)_2$, then duty cycle of the waveform generated at Q_1 output 66.67 %.



$Q_1 \rightarrow T_{off} = 1T_{clk}$

$T_{ON} = 2T_{clk}$

$T_{Q_1} = 3T_{clk}$

$f_{Q_1} = f_{clk}/3$

$\%\ \text{duty cycle} = \frac{2T_{clk}}{3T_{clk}} \times 100\%$

$= 66.67\%$

Shift Register

For what purpose shift registers are used ?

→ to store the data.

- Types of shift registers :

→ SI SO

→ SI PO

→ PI PO

→ PI SO



Topic : 2 Min Summary

Synch. Counter

Thank you

GW

Soldiers !

